1 Dynamic Response Analysis Using Newmark-β Method

1.1 Problem Definition

A linear single-degree-of-freedom (SDOF) system is defined by:

$$m = 3600 \text{ kg} \tag{1}$$
$$k = 1400 \text{ kN/m} = 1.4 \times 10^6 \text{ N/m} \tag{2}$$
$$c = 70 \text{ kN·s/m} = 7.0 \times 10^4 \text{ N·s/m} \tag{3}$$

The system is subjected to a time-varying external force $F(t)$ as shown in the assignment. The equation of motion is:

$$m\ddot{u}(t) + c\dot{u}(t) + ku(t) = F(t) \tag{4}$$

The system is initially at rest:

$$u(0) = 0, \quad \dot{u}(0) = 0 \tag{5}$$

—

1.2 Newmark-β Method

The Newmark-β method with constant average acceleration is used:

$$\beta = \frac{1}{4}, \quad \gamma = \frac{1}{2} \tag{6}$$

For the constant average acceleration method ($\beta = 1/4$, $\gamma = 1/2$), the following constants are defined:

$$a_0 = \frac{1}{\beta \Delta t^2}, \qquad a_1 = \frac{\gamma}{\beta \Delta t}, \tag{7}$$
$$a_2 = \frac{1}{\beta \Delta t}, \qquad a_3 = \frac{1}{2\beta} - 1, \tag{8}$$
$$a_4 = \frac{\gamma}{\beta} - 1, \qquad a_5 = \Delta t \left(\frac{\gamma}{2\beta} - 1 \right) \tag{9}$$

For $\Delta t = 0.1$ s:

$$a_0 = 400, \quad a_1 = 20, \quad a_2 = 40, \quad a_3 = 1, \quad a_4 = 1, \quad a_5 = 0 \tag{10}$$

The method leads to the following recurrence relations:

$$u_{i+1} = \frac{\hat{F}_{i+1}}{k_{\text{eff}}} \tag{11}$$

$$\ddot{u}_{i+1} = a_0(u_{i+1} - u_i) - a_2\dot{u}_i - a_3\ddot{u}_i \tag{12}$$

$$\dot{u}_{i+1} = \dot{u}_i + \Delta t\left[(1-\gamma)\ddot{u}_i + \gamma\ddot{u}_{i+1}\right] \tag{13}$$

where the effective force is:

$$\hat{F}_{i+1} = F_{i+1} + m(a_0u_i + a_2\dot{u}_i + a_3\ddot{u}_i) + c(a_1u_i + a_4\dot{u}_i + a_5\ddot{u}_i) \tag{14}$$

and the effective stiffness is:

$$k_{\text{eff}} = k + a_0m + a_1c \tag{15}$$

For $\Delta t = 0.1$ s,

$$k_{\text{eff}} = 1.4 \times 10^6 + 400(3600) + 20(70000) = 4.24 \times 10^6 \text{ N/m} \tag{16}$$

—

1.3 Time Integration Parameters

$$\Delta t = 0.1 \text{ s}, \quad t_{\text{end}} = 2.0 \text{ s} \tag{17}$$

The loading is applied for $t \leq 1.0$ s and set to zero afterward, resulting in free vibration beyond this point.

—

1.4 Numerical Implementation

The response was computed using a Python implementation of the Newmark-β method. The algorithm evaluates displacement, velocity, and acceleration at each time step. An Excel file was also used to verify and validate the results.

—

1.5 Results

The response was computed for two time steps, $\Delta t = 0.1$ s and $\Delta t = 0.01$ s. Requested tabulated outputs are provided below for time, force, displacement, velocity, acceleration, and effective force.

As a consistency check, the maximum static displacement is estimated as:

$$u_{\text{static}} = \frac{F_{\max}}{k} = \frac{40000}{1.4 \times 10^6} \approx 2.86 \times 10^{-2} \text{ m} \tag{18}$$

The computed maximum dynamic displacement is approximately:

$$u_{\max} \approx 3.06 \times 10^{-2} \text{ m} \tag{19}$$

The dynamic displacement is slightly larger than the static estimate, which is expected due to dynamic amplification effects.

The natural frequency of the system is:

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{1.4 \times 10^6}{3600}} \approx 19.7 \text{ rad/s} \tag{20}$$

The corresponding natural period is:

$$T_n = \frac{2\pi}{\omega_n} \approx 0.319 \text{ s} \tag{21}$$

The oscillation period observed in the response plots is consistent with this value.

1.5.1 Table for $\Delta t = 0.1$ s

Table 1: Response history for $\Delta t = 0.1$ s

t (s)	F (N)	u (m)	$\dot{u}$ (m/s)	$\ddot{u}$ (m/s^2)	$\hat{F}$ (N)
0.00	0	0.000000e+00	0.000000e+00	0.000000e+00	0.00
0.10	8000	1.886792e-03	3.773585e-02	7.547170e-01	8000.00
0.20	18000	8.054468e-03	8.561766e-02	2.029192e-01	34150.94
0.30	36000	1.837910e-02	1.208750e-01	5.022267e-01	77927.38
0.40	39000	2.803582e-02	7.225957e-02	-1.474534e+00	118871.89
0.50	40000	3.060777e-02	-2.082056e-02	-3.870682e-01	129776.96
0.60	39000	2.832006e-02	-2.493383e-02	3.048028e-01	120077.04
0.70	31000	2.528076e-02	-3.585215e-02	-5.231692e-01	107190.41
0.80	19000	1.916075e-02	-8.654799e-02	-4.907475e-01	81241.58
0.90	10000	1.040768e-02	-8.851338e-02	4.514396e-01	44128.57
1.00	0	2.887060e-03	-6.189905e-02	8.084708e-02	12241.13
1.10	0	-1.121721e-03	-1.827657e-02	7.916025e-01	-4756.10
1.20	0	-1.001675e-03	2.067748e-02	-1.252160e-02	-4247.10
1.30	0	3.620623e-04	6.597278e-03	-2.690824e-01	1535.14
1.40	0	3.470230e-04	-6.898063e-03	-8.244064e-04	1471.38
1.50	0	-1.164169e-04	-2.370737e-03	9.137092e-02	-493.61
1.60	0	-1.200534e-04	2.298007e-03	2.003967e-03	-509.03
1.70	0	3.727267e-05	8.485146e-04	-3.099382e-02	158.04
1.80	0	4.147612e-05	-7.644456e-04	-1.265382e-03	175.86
1.90	0	-1.187608e-05	-3.025983e-04	1.050233e-02	-50.35
2.00	0	-1.431031e-05	2.539136e-04	6.279094e-04	-60.68

1.5.2 Standalone Plots for $\Delta t = 0.1$ s

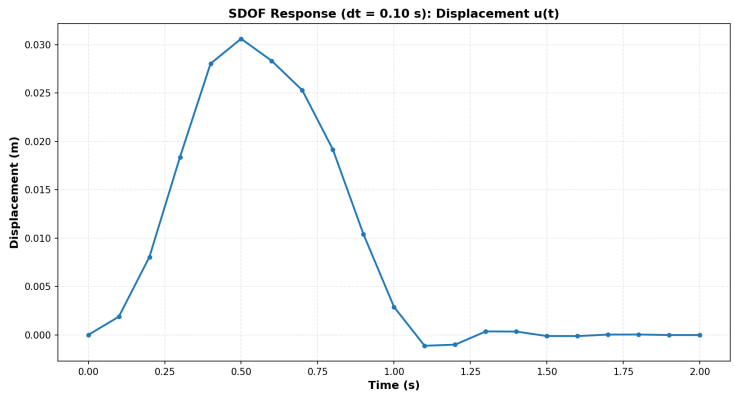

Figure 1: Displacement response for $\Delta t = 0.1$ s

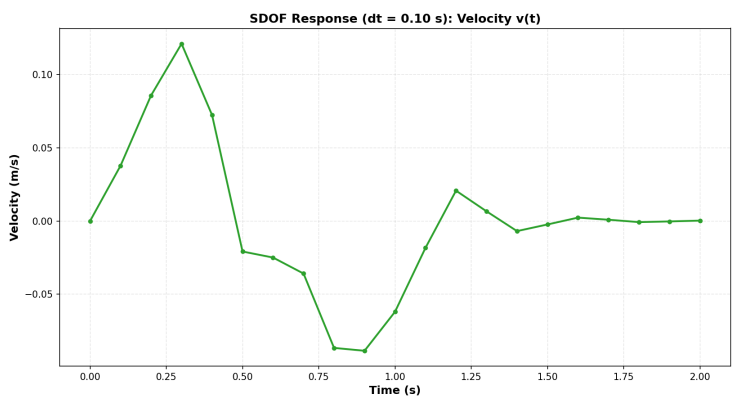

Figure 2: Velocity response for $\Delta t = 0.1$ s

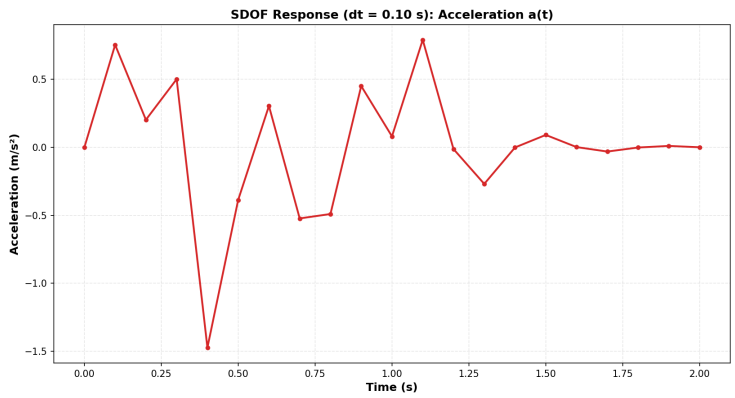

Figure 3: Acceleration response for $\Delta t = 0.1$ s

1.5.3 Table for $\Delta t = 0.01$ s

Table 2: Sampled response history for $\Delta t = 0.01$ s

t (s)	F (N)	u (m)	$\dot{u}$ (m/s)	$\ddot{u}$ (m/s^2)	$\hat{F}$ (N)
0.00	0	0.000000e+00	0.000000e+00	0.000000e+00	0.00
0.06	4800	5.747266e-04	2.510103e-02	6.217529e-01	91611.42
0.12	10000	3.112619e-03	5.752280e-02	4.488160e-01	496151.42
0.18	16000	7.210806e-03	7.618853e-02	1.587986e-01	1149402.42
0.24	25200	1.210734e-02	9.170017e-02	5.085324e-01	1929909.45
0.30	36000	1.855460e-02	1.221247e-01	4.096757e-01	2957603.00
0.36	37800	2.533206e-02	8.935438e-02	-1.088801e+00	4037929.66
0.42	39200	2.877302e-02	2.810243e-02	-8.470568e-01	4586420.11
0.48	39800	2.929157e-02	-4.804142e-03	-2.421986e-01	4669077.05
0.54	39600	2.880148e-02	-9.622536e-03	-1.347094e-02	4590956.01
0.60	39000	2.823385e-02	-8.910194e-03	2.675564e-02	4500476.06
0.66	34200	2.724435e-02	-2.934867e-02	-5.243572e-01	4342749.72
0.72	28600	2.459707e-02	-5.813043e-02	-4.907698e-01	3920773.35
0.78	21400	2.027747e-02	-8.397833e-02	-3.083259e-01	3232228.51
0.84	15400	1.494569e-02	-8.843267e-02	1.850903e-01	2382342.50
0.90	10000	1.009166e-02	-7.264706e-02	2.658258e-01	1608610.31
1.00	0	3.357203e-03	-6.282655e-02	2.129504e-01	535194.31
1.20	0	-4.007250e-04	1.082452e-02	-5.463922e-02	-63875.57
1.40	0	2.852457e-05	-1.320617e-03	1.698307e-02	4548.22
1.60	0	7.775912e-06	2.613269e-04	-3.373681e-03	1239.21
1.80	0	-1.501355e-07	-2.812460e-05	6.052533e-04	-23.93
2.00	0	9.216740e-08	3.105826e-06	-9.623394e-05	14.69

1.5.4 Comparison Plots: $\Delta t = 0.01$ s vs $\Delta t = 0.1$ s

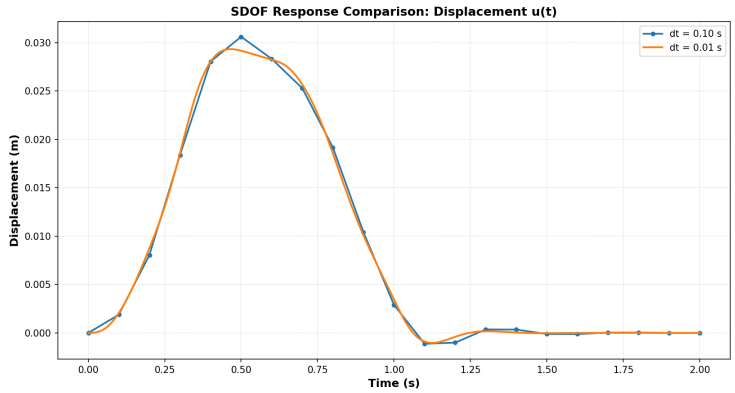

Figure 4: Displacement comparison between $\Delta t = 0.01$ s and $\Delta t = 0.1$ s

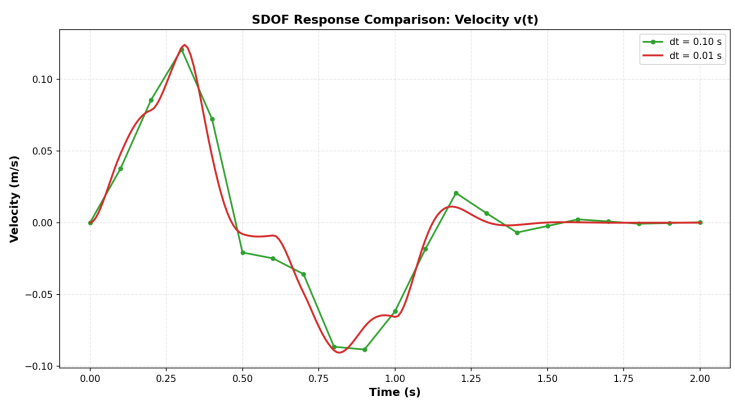

Figure 5: Velocity comparison between $\Delta t = 0.01$ s and $\Delta t = 0.1$ s

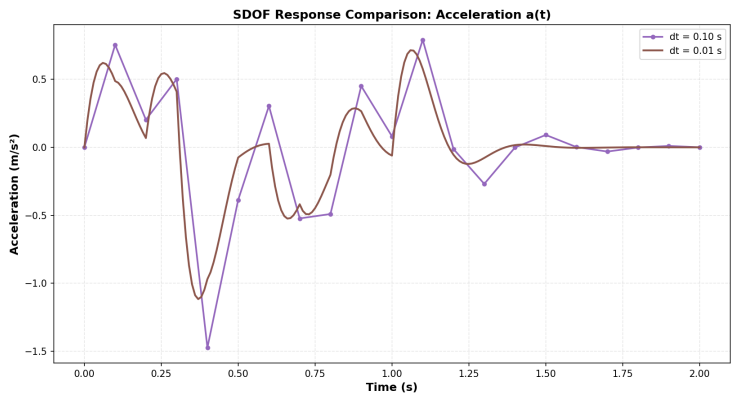

Figure 6: Acceleration comparison between $\Delta t = 0.01$ s and $\Delta t = 0.1$ s

—

1.6 Discussion

The response of the system consists of two distinct phases: forced vibration for $0 \leq t \leq 1.0$ s and free vibration for $t > 1.0$ s, after the external loading is removed.

During the forced vibration phase, the displacement increases as the applied force rises, reaching a peak value of approximately 3.06×10^{-2} m at $t \approx 0.5$ s. This value is slightly higher than the corresponding static displacement estimate,

$$u_{\text{static}} = \frac{F_{\max}}{k} \approx 2.86 \times 10^{-2} \text{ m},$$

indicating the presence of dynamic amplification due to inertia effects.

The response exhibits oscillatory behavior consistent with the natural period of the system, $T_n \approx 0.319$ s. The displacement, velocity, and acceleration plots all reflect this periodic behavior, confirming that the numerical solution captures the fundamental dynamic characteristics of the system.

After $t = 1.0$ s, the system undergoes free vibration. Due to the presence of damping ($c = 7.0 \times 10^4$ N·s/m), the amplitude of oscillation gradually decays over time. By $t = 2.0$ s, the response has nearly diminished, with the displacement approaching zero and only small residual oscillations remaining.

The comparison between $\Delta t = 0.1$ s and $\Delta t = 0.01$ s shows that both time steps produce consistent peak values and overall trends. However, the smaller time step yields a smoother and more refined response, particularly in the velocity and acceleration histories. This indicates improved numerical resolution of the dynamic response.

Overall, the results demonstrate that the Newmark-β method with constant average acceleration provides stable and accurate solutions for the given time-step sizes. The close agreement between the two time steps suggests that $\Delta t =$

0.1 s is sufficient for capturing the essential behavior of the system, while $\Delta t =$ 0.01 s offers higher fidelity at increased computational cost.

2 Elasto-Plastic Dynamic Response Analysis

2.1 Problem Definition

The system considered in Question 1 is now assumed to exhibit a perfectly elasto-plastic force-displacement relationship with yield force:

$$F_y = 24 \text{ kN} = 24000 \text{ N} \tag{22}$$

The equation of motion becomes:

$$m\ddot{u}(t) + c\dot{u}(t) + F_r(t) = F(t) \tag{23}$$

where $F_r(t)$ is the nonlinear restoring force.

2.2 Elasto-Plastic Constitutive Model

The restoring force is modeled as perfectly elasto-plastic in terms of elastic and plastic displacement components:

$$u = u_e + u_p, \qquad F_r = ku_e = k(u - u_p) \tag{24}$$

At each step, a trial restoring force is formed from the previous plastic displacement:

$$F_{\text{trial}} = k\left(u_{n+1}^{\text{trial}} - u_{p,n}\right) \tag{25}$$

The return-mapping rule is:

$$F_{r,n+1} = \begin{cases} k\left(u_{n+1} - u_{p,n}\right), & |F_{\text{trial}}| \leq F_y \\ \pm F_y, & |F_{\text{trial}}| > F_y \end{cases} \tag{26}$$

with plastic displacement update in the yielded state:

$$u_{p,n+1} = u_{n+1} - \frac{F_{r,n+1}}{k} \tag{27}$$

where:

- u_p is the plastic displacement,
- k is the elastic stiffness,
- F_y is the yield force.

The yield displacement is:

$$u_y = \frac{F_y}{k} \tag{28}$$

2.3 Numerical Solution Procedure

The nonlinear dynamic response is computed using a step-by-step incremental direct integration algorithm, as illustrated in Figure X. The method advances the solution in time without iteration by updating the system state at each time step based on the tangent stiffness and equilibrium conditions.

The governing equation is:

$$m\ddot{u} + c\dot{u} + F_s(u) = F(t) \tag{29}$$

First, the initial acceleration is obtained from equilibrium:

$$\ddot{u}_0 = \frac{1}{m}\left(F_0 - c\dot{u}_0 - F_s(u_0)\right) \tag{30}$$

At each time step i, the following procedure is applied:

1. Compute the effective stiffness:

$$k_i^* = k_i(u_i) + \frac{2c}{\Delta t} + \frac{4m}{\Delta t^2} \tag{31}$$

2. Compute the effective force increment:

$$\Delta F_i^* = \Delta F_i + \left(\frac{4m}{\Delta t} + 2c\right)\dot{u}_i + 2m\ddot{u}_i \tag{32}$$

3. Solve for the displacement increment:

$$\Delta u_i = \frac{\Delta F_i^*}{k_i^*} \tag{33}$$

4. Compute velocity and acceleration increments:

$$\Delta\dot{u}_i = \frac{2}{\Delta t}\Delta u_i - 2\dot{u}_i \tag{34}$$

$$\Delta\ddot{u}_i = \frac{4}{\Delta t^2}\Delta u_i - \frac{4}{\Delta t}\dot{u}_i - 2\ddot{u}_i \tag{35}$$

5. Update displacement and velocity:

$$u_{i+1} = u_i + \Delta u_i \tag{36}$$

$$\dot{u}_{i+1} = \dot{u}_i + \Delta\dot{u}_i \tag{37}$$

6. Evaluate the restoring force $F_{s,i+1}$ using the elasto-plastic constitutive model and update the plastic displacement.

7. Compute the acceleration from dynamic equilibrium:

$$\ddot{u}_{i+1} = \frac{F_{i+1} - c\dot{u}_{i+1} - F_{s,i+1}}{m} \tag{38}$$

8. Advance to the next time step and repeat.

This formulation allows the nonlinear response to be obtained efficiently without iterative correction within each time step, while maintaining consistency with the governing dynamic equilibrium.

2.4 Results

The displacement, velocity, and acceleration time histories for both linear and elasto-plastic cases are shown below.

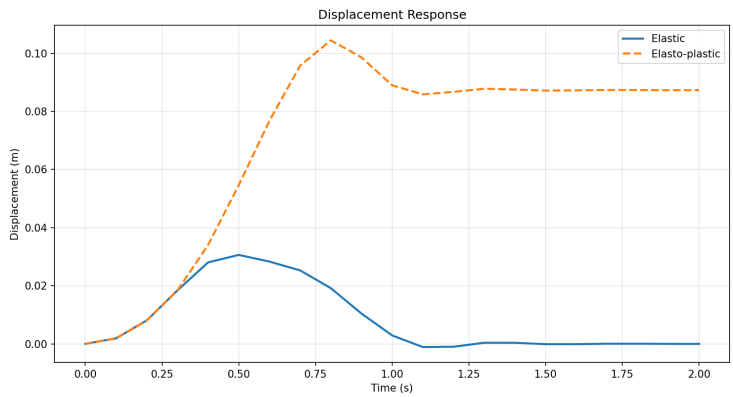

Figure 7: Displacement response comparison

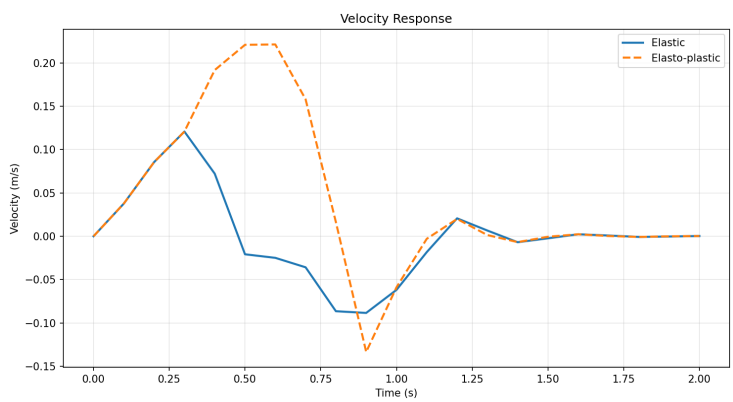

Figure 8: Velocity response comparison

2.5 Force–Displacement Relationship

The force-displacement relationship for both cases is shown in Figure below. The elastic reference line is drawn only up to the yield force level ($\pm F_y$) for a consistent comparison with the elasto-plastic response.

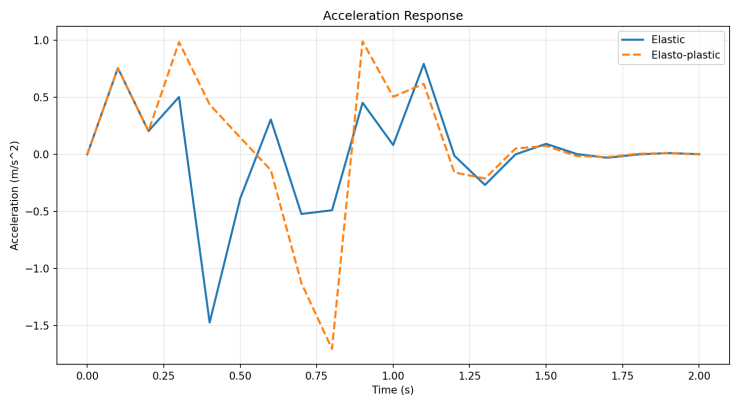

Figure 9: Acceleration response comparison

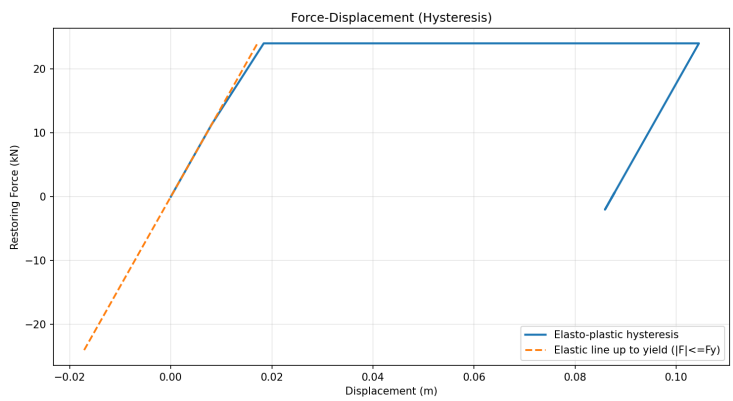

Figure 10: Force-displacement relationship (elastic vs elasto-plastic)

2.6 Discussion

The elasto-plastic system exhibits significantly different behavior compared to the linear-elastic system.

- **Increased displacement:** After yielding, the effective stiffness reduces, leading to larger displacements.
- **Force limitation:** The restoring force is capped at $\pm F_y$, preventing further force increase.
- **Energy dissipation:** Plastic deformation dissipates energy, reducing oscillation amplitudes.
- **Residual displacement:** Unlike the elastic system, the elasto-plastic system does not return to zero displacement after loading ceases.

The hysteresis loop clearly illustrates the nonlinear behavior and energy dissipation mechanism.

2.7 Summary of Results

- Maximum displacement in the elasto-plastic system is greater than that of the elastic system.
- The restoring force remains bounded within $\pm F_y$.
- Residual displacement is observed at the end of the simulation.
- Equilibrium residuals after convergence are near numerical precision.

2.8 Latest Script Output (Incremental Direct Integration)

The nonlinear solver was executed with the CE586 incremental direct integration algorithm and the following outputs were obtained:

- Yield displacement: $u_y = F_y/k = 1.714286 \times 10^{-2}$ m
- Restoring force cap check: $\max|F_s| = 24000.000000$ N ≤ 24000.000001 N (Pass)
- Peak elastic displacement: $\max|u_{\text{elastic}}| = 3.060777 \times 10^{-2}$ m
- Peak inelastic displacement: $\max|u_{\text{inelastic}}| = 1.044710 \times 10^{-1}$ m
- Distinct response check: $\max|u_{\text{inelastic}} - u_{\text{elastic}}| = 8.820017 \times 10^{-2}$ m (Pass)
- Residual displacement at $t = 2.0\,\text{s}$: 8.731893×10^{-2} m
- Max restoring force: 24000.00 N
- Max equilibrium residual:

$$\max|F - ma - cv - F_s| = 1.819 \times 10^{-12} \text{ N}$$